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Activation of crystalline cellulose to cellulose III_I results in efficient hydrolysis by cellobiohydrolase

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The crystalline polymorphic form of cellulose (cellulose I_α-rich) of the green alga, *Cladophora*, was converted into cellulose III_I and I_β by supercritical ammonium and hydrothermal treatments, respectively, and the hydrolytic rate and the adsorption of *Trichoderma viride* cellobiohydrolase I (Cel7A) on these products were evaluated by a novel analysis based on the surface density of the enzyme. Cellobiose production from cellulose III_I was more than 5 times higher than that from cellulose I. However, the amount of enzyme adsorbed on cellulose III_I was less than twice that on cellulose I, and the specific activity of the adsorbed enzyme for cellulose III_I was more than 3 times higher than that for cellulose I. When cellulose III_I was converted into cellulose I_β by hydrothermal treatment, cellobiose production was dramatically decreased, although no significant change was observed in enzyme adsorption. This clearly indicates that the enhanced hydrolysis of cellulose III_I is related to the structure of the crystalline polymorph. Thus, supercritical ammonium treatment activates crystalline cellulose for hydrolysis by cellobiohydrolase.

Cellulose is a linear polymer of β-1,4-linked anhydrous glucose residues, and is the major component of plant cell walls. In nature, cellulose chains are packed into ordered arrays to form insoluble microfibrils, which are stabilized by cross-links involving intermolecular hydrogen bonds. Microfibrils generally consist of a mixture of disordered amorphous cellulose and cellulose I, which forms highly ordered crystalline regions. Cellulose I is further classified into two polymorphs, triclinic cellulose I_α, which is found in algal and bacterial celluloses, and monoclinic cellulose $I_{\beta,}$ called cotton-ramie-type cellulose [1–3]. Although the differences in their physiological roles in the cell wall are uncertain, cellulose I_α is more susceptible than cellulose I_β to hydrolysis by cellulase [4,5].

Cellulase is a generic term for enzymes hydrolyzing β-1,4-glucosidic linkages. If we consider the structure of microfibrils, however, cellulases should be subdivided into two categories, as all cellulases can hydrolyze amorphous cellulose, whereas only a limited number can hydrolyze crystalline cellulose [6]. The enzymes that hydrolyze crystalline cellulose are generally called cellobiohydrolases, and share similar two-domain structures, with a catalytic domain (CD) and a cellulose-binding domain (CBD) [7–10]. As the initial step of the reaction, they are adsorbed on the surface of crystalline cellulose via the CBD, then glucosidic linkages are hydrolyzed by the CD. As the reaction produces mainly cellobiose, a soluble β-1,4-glucosidic dimer, from insoluble substrates, the hydrolysis of crystalline cellulose occurs at a solid/liquid interface [11–13]. To evaluate such reactions, we recently developed a novel analysis based on surface density (ρ), defined as the amount of adsorbed enzyme (A) divided by the maximum adsorption of the enzyme ($A_{\max}$) [14]. Using this parameter, we were able to analyze the hydrolysis of crystalline cellulose while taking account of the available substrate

Abbreviations
CBD, cellulose-binding domain; CD, catalytic domain; FT-IR: Fourier-transform infrared.

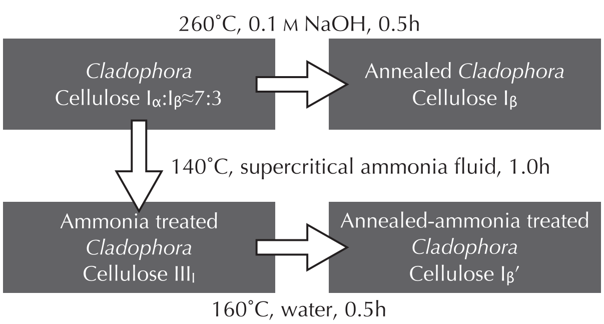
surface area, which is not only dependent on the origin of the cellulose, but also changes during hydrolysis. The results showed that the higher hydrolytic rate of cellulose I_α than cellulose I_β is due to the difference in crystal structure, but not to the difference in surface area accessible to cellulase [14].

Cellulose III_I, which is the designation given to ammonia-treated cellulose, is a reactive crystalline cellulose which is used as a precursor of many cellulose derivatives [15,16]. Wada and coauthors [17] solved the crystal structure of cellulose III_I by synchrotron X-ray and neutron fiber diffraction analyses, and showed that it has a lower packing density than cellulose I_α or I_β. In this study, we analyzed the hydrolysis of cellulose III_I by cellobiohydrolase in terms of surface density, and discuss how the structural differences of crystalline celluloses affect the hydrolytic activity of cellobiohydrolase.

Results

Cellulose preparations

Different crystalline polymorphs of *Cladophora* cellulose (I_α-rich) were prepared as shown in Scheme 1. Figure 1 shows the Fourier-transform infrared (FT-IR) spectra of the OH stretching region for the samples. The absorption band at 3240 cm^{-1}, which is assigned to cellulose I_α, is seen in the spectrum of the native *Cladophora* cellulose (Fig. 1A), whereas the hydrothermal-treated celluloses had a band at 3270 cm^{-1} (Fig. 1B,D) without that at 3240 cm^{-1}, suggesting that they have all been converted into cellulose I_β. The sharp band at 3480 cm^{-1} in Fig. 1C indicates that cellulose I was completely converted into cellulose III_I by the supercritical ammonia treatment. The cellulose III_I was further converted into cellulose I_β by subsequent hydrothermal treatment, as indicated by similar FT-IR spectra in Fig. 1B,D.



Scheme 1. Conversion of crystalline polymorphs of *Cladophora* cellulose.

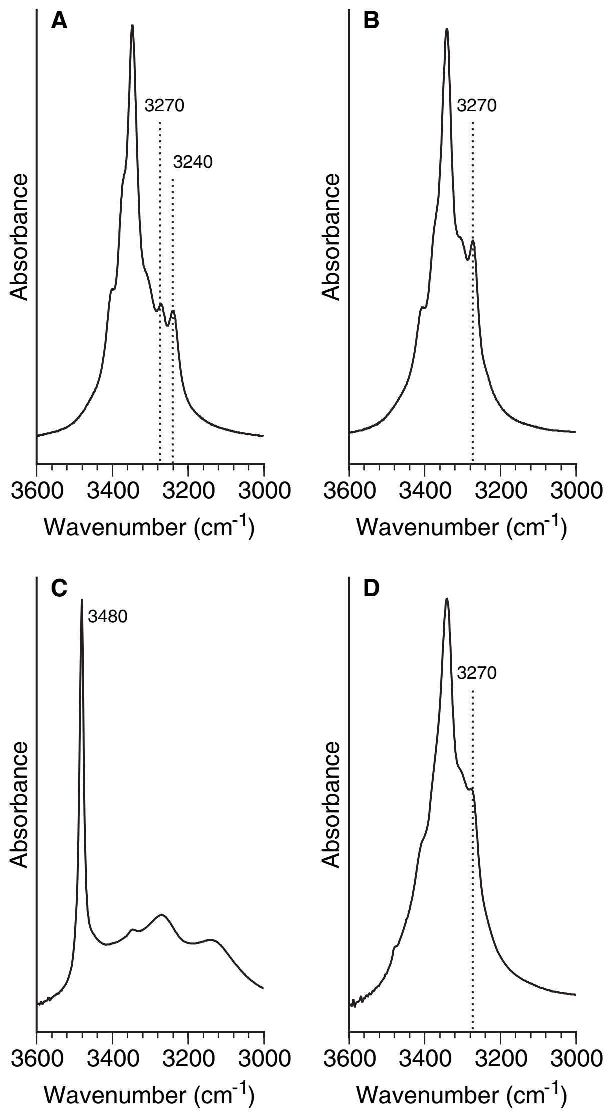


Fig. 1. FT-IR spectra of highly crystalline celluloses in the OH stretching regions. (A) Native *Cladophora* cellulose; (B) hydrothermal treated cellulose; (C) supercritical ammonia-treated cellulose; (D) supercritical ammonia and hydrothermal treated cellulose. Bands at 3240 and 3270 cm^{-1} are assigned to the cellulose I_α and I_β phase [36], respectively.

Hydrolysis of crystalline celluloses and adsorption of Cel7A

The time course of increase in cellobiose concentration during cellulose hydrolysis, measured using the cellobiose dehydrogenase–cytochrome *c* redox system, is shown in Fig. 2. Although apparent differences in cellobiose production among cellulose I samples were observed, the most dramatic increase in hydrolysis by Cel7A was obtained after conversion of the samples

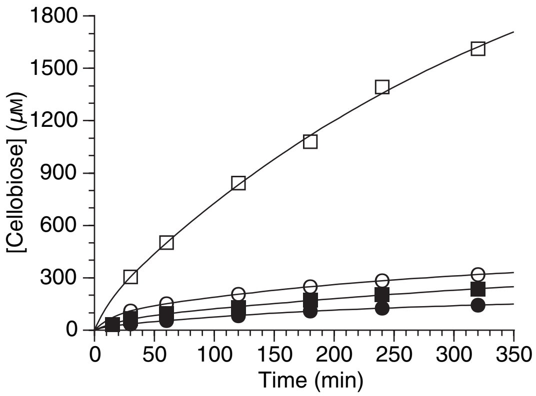


Fig. 2. Time course of cellobiose concentration in the reaction mixtures of highly crystalline celluloses with Cel7A. ■, Cellulose I_α-rich; ●, cellulose I_β; □, cellulose III_I; ○, cellulose $I_\beta{}'$. Highly crystalline cellulose (0.1%, w/v) was incubated with 2.2 μM Cel7A in 50 mM sodium acetate (pH 5.0) at 30 °C. Cellobiose concentration in the supernatant after termination of the reaction by centrifugation (twice at 15 000 ***g*** for 5 min) was determined with the cellobiose dehydrogenase–cytochrome *c* redox system as described [14].

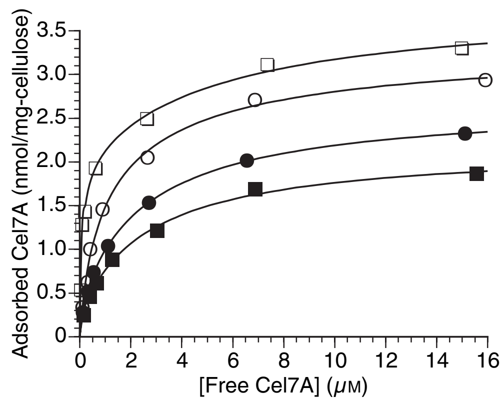


Fig. 3. Enzyme concentration dependence of the amount of adsorbed Cel7A. ■, Cellulose I_α-rich; ●, cellulose I_β; □, cellulose III_I; ○, cellulose $I_\beta{}'$. Cel7A was incubated with 1 mg·mL^{-1} crystalline cellulose In 1 mL 50 mM sodium acetate, pH 5.0, at 30 °C. This figure shows adsorption of Cel7A after incubation for 120 min as representative results of four time points (120, 180, 240, and 320 min). The lines indicate the fitting of the data to the two-binding-site model.

into cellulose III_I by supercritical ammonia treatment. The cellobiose concentration produced from cellulose III_I was 1600 μM after 320 min incubation and degradation reached 50% of the initial substrate, whereas the extent of hydrolysis of other cellulose samples was less than 10%, demonstrating that the hydrolyzability of crystalline cellulose is dramatically activated if the crystalline polymorphic form is converted into cellulose III_I.

Adsorption of Cel7A on the crystalline cellulose samples was examined, and the data were fitted to the two-binding-site Langmuir model as shown in Fig. 3. Ammonia treatment might increase the surface area available to the enzyme, as the amounts of adsorbed enzyme on cellulose III_I and cellulose $I_\beta{}'$ were 1.5–2 times higher than on the samples without ammonia treatment (cellulose I_α-rich and I_β). The adsorption parameters (K_{ad1}, K_{ad2}, A_1, A_2, A_{max}, $A_1{\cdot}K_{ad1}$, and $A_2{\cdot}K_{ad2}$) listed in Table 1 show that the difference made by ammonia treatment was mainly due to differences in A_1, the maximum adsorption of high-affinity binding: A_1 for cellulose III_I was almost 8 times higher than that for cellulose I_α-rich substrate, and A_1 for cellulose $I_\beta{}'$ was 2.6 times that for cellulose I_β, although no significant difference was observed in A_2 among the four crystalline cellulose samples. In addition, the K_{ad1} value of Cel7A on cellulose III_I was quite high compared with those on other celluloses. These result in a higher adsorption efficiency ($A_1{\cdot}K_{ad1}$) on cellulose III_I compared with other crystalline cellulose I samples.

Surface density analysis of the hydrolysis of crystalline cellulose

Figure 4 shows the surface density (ρ) dependence of cellobiose production rate (*v*) from crystalline celluloses. As expected from Fig. 2, the highest hydrolytic rate by Cel7A was seen with cellulose III_I. When cellulose I samples were used as substrates, the maximum *v* values were observed at ρ = 0.3–0.4, whereas, in the case of cellulose III_I, the maximum rate (5.3 μM·min^{-1}) was achieved at a surface density of 0.55. This means that empty space on the substrate surface equivalent to another 2 enzyme molecules per adsorbed molecule must be left on cellulose I to achieve maximum hydrolysis, whereas empty space equivalent to only 1 molecule is enough on cellulose III_I.

The specific activity of adsorbed enzyme ($k = v/A$) towards crystalline cellulose samples was plotted against surface density as shown in Fig. 5. The *k* values of all samples declined linearly with increase in ρ when a logarithmic scale was used for the *y*-axis. The calculated values of *k* at ρ → 0 (k_0) and reduction rate of *k* (*B*) are listed in Table 2. The k_0 for cellulose III_I was approximately 3 times higher than those for cellulose I samples. Moreover, the *B* value for cellulose III_I is very much lower than those for cellulose I. These results indicate that the reason for the higher rate of hydrolysis of cellulose III_I by Cel7A is the higher specific activity of the enzyme for this crystalline polymorph, not the larger surface area of the substrate.

Table 1. Adsorption parameters of Cel7A for highly crystalline celluloses. The adsorption parameters were calculated by nonlinear fitting of the data after incubation in 50 mM sodium acetate, pH 5.0, for 120, 180, 240, and 320 min. K_{ad1} and K_{ad2} are expressed as μM^{-1}, A_1, A_2, and A_{max} as nmol·(mg cellulose)$^{-1}$, and $A_1 \cdot K_{ad1}$ and $A_2 \cdot K_{ad2}$ as ml·(mg cellulose)$^{-1}$.

	K_{ad1}	K_{ad2}	A_1	A_2	A_{max}	$A_1 \cdot K_{ad1}$	$A_2 \cdot K_{ad2}$
Cellulose I_α-rich	8.5 ± 0.7	0.44 ± 0.04	0.22 ± 0.02	2.0 ± 0.2	2.2 ± 0.2	1.9	0.88
Cellulose I_β	4.7 ± 0.4	0.43 ± 0.04	0.58 ± 0.03	2.1 ± 0.3	2.6 ± 0.3	2.7	0.90
Cellulose III_I	13 ± 2	0.27 ± 0.04	1.8 ± 0.5	1.8 ± 0.5	3.7 ± 1.0	23	0.49
Cellulose I_β'	2.5 ± 0.5	0.22 ± 0.02	1.5 ± 0.1	2.0 ± 0.5	3.5 ± 0.6	3.7	0.44

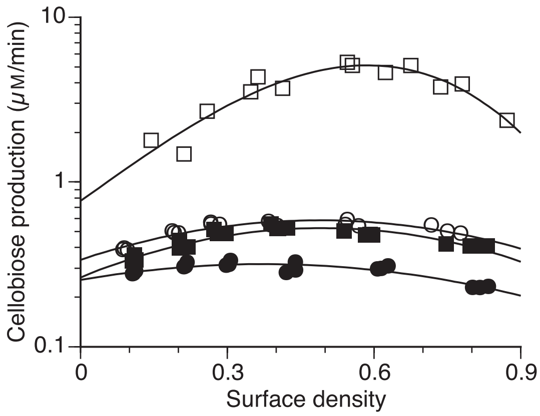


Fig. 4. Surface density (ρ) dependence of cellobiose production (v) from crystalline celluloses. ■, Cellulose I_α-rich; ●, cellulose I_β; □, cellulose III_I; ○, cellulose I_β'. The plots were obtained from the results after incubation for 120, 180, 240, and 320 min.

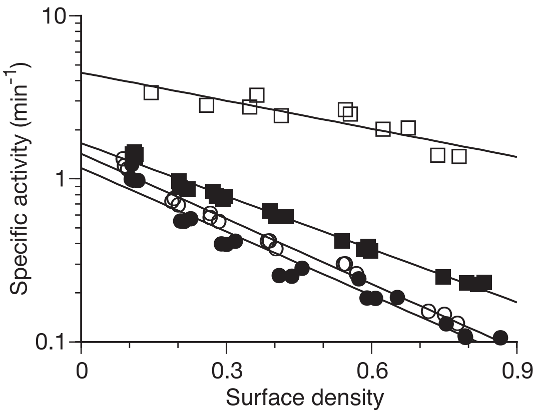


Fig. 5. Surface density (ρ) dependence of specific activity of adsorbed Cel7A (k). ■, Cellulose I_α-rich; ●, cellulose I_β; □, cellulose III_I; ○, cellulose I_β'. The plots were obtained from the results after incubation for 120, 180, 240, and 320 min. The ρ and k values were estimated as reported previously [14].

Discussion

In order to utilize cellulosic biomass for bioethanol production or biorefining, effective hydrolysis of crystalline cellulose is critical, because ≈70% of natural

Table 2. The k value at $\rho \rightarrow 0$ (k_0) and reduction rate of k (B) for hydrolysis of crystalline celluloses. These parameters were calculated from ρ–k plots in Fig. 5 using Eqn (1) as described in Experimental procedures.

	k_0 (min^{-1})	B
Cellulose I_α-rich	1.7 ± 0.1	2.5 ± 0.2
Cellulose I_β	1.2 ± 0.1	3.0 ± 0.2
Cellulose III_I	4.5 ± 0.4	1.3 ± 0.1
Cellulose I_β'	1.4 ± 0.1	3.1 ± 0.2

cellulose is crystalline. However, the rate of degradation of cellulose I by cellulase is extremely low compared with that of amorphous cellulose, possibly because of its tightly packed structure [6]. There are many pretreatment methods to enhance the hydrolyzability of cellulosic biomass, and they generally include a step for disrupting the crystal structure by physical and/or chemical treatment. Among them, ammonia treatment is a simple and effective method [15,16]. In the present study, we used our surface density analysis to analyze the enhanced hydrolysis of crystalline cellulose following ammonia treatment, which converts cellulose I into cellulose III_I, and we show that cellulose III_I is an intrinsically activated form of celluose, which is highly susceptible to hydrolysis.

The adsorption of cellobiohydrolase on crystalline cellulose is well described by a two-binding-site model [13], and we proposed that the high-affinity and low-affinity adsorption can be interpreted as productive and nonproductive binding, respectively, based on the two-domain structure of cellobiohydrolase and the ρ dependence of cellobiose production [14]. In that study, we mainly focused on the K_{ad1} values to explain the different hydrolytic rates of cellulose I_α and I_β. However, the efficiency of high-affinity adsorption ($A_1 \cdot K_{ad1}$) may also affect the activity when we evaluate total cellobiose production in the reaction mixture, as this value resembles catalytic efficiency (V_{max}/K_m) in the Michaelis–Menten model when Cel7A is productively bound on the surface of cellulose. A comparison of adsorption parameters (Table 1) and cellobiose production (Fig. 4) in the present study suggests that the $A_1 \cdot K_{ad1}$ values correlate with cellobiose production, as larger $A_1 \cdot K_{ad1}$ values are associated with greater cellobiose production from cellulose III_I. Although it is still difficult to interpret the results quantitatively, all our results are consistent with a correlation between high-affinity adsorption and cellobiose production. The three-dimensional structures of the CD and CBD of *Trichoderma* Cel7A showed that this enzyme accommodates at least 10 glucose residues at the active-site tunnel of the CD [18,19], whereas CBD binds to the cellulose surface via hydrophobic interaction between three tyrosine residues and glucose residues [20,21]. Therefore, it is reasonable that the productive binding by both CD and CBD would involve very much higher affinity than nonproductive binding, in which only the CBD contributes to the adsorption. As far as we know, the results observed in this study represent the first evidence that the putative productive binding mode is truly productive.

We previously reported that the specific activity of adsorbed enzyme (k) is greatly influenced by the crystalline polymorphic form of the substrate. Moreover, in the cases of cellulose I_β from *Halocynthia* and hydrothermally treated *Cladophora*, similar ρ–k plots should be obtained if crystalline celluloses with the same polymorphic form are used as substrates, because the ρ value is independent of the surface area of each sample. In the present study, although the rate of cellobiose production from cellulose I_β' is higher than that from cellulose I_β (Fig. 4), the specific activity of the adsorbed enzyme (Fig. 5) was almost the same with cellulose I_β and I_β'. These results can be interpreted as indicating that the reason for the higher cellobiose production from cellulose I_β' than cellulose I_β is the larger amount of adsorption during hydrolysis, but not an increase in specific activity. There are several studies showing that conversion into cellulose III_I decreases the crystal size [22,23]. Therefore, treatment with supercritical ammonia increases the surface area available for cellobiohydrolase (possibly the hydrophobic surface) and thus increases the number of enzyme molecules that can be adsorbed on the surface of cellulose III_I and I_β' (Fig. 3 and Table 1).

The recent synchrotron X-ray and neutron fiber diffraction studies of crystalline celluloses [17,24,25] have shown that cellulose III_I has a one-chain monoclinic unit cell with an asymmetric unit containing only one glucosyl residue, and this is quite different from cellulose I. The views from the hydrophobic surface and from the nonreducing end of each chain are compared among cellulose I_α, cellulose I_β, and cellulose III_I in Fig. 6. The structure of cellulose III_I results in a lower packing density than that of cellulose I, with a greater distance between hydrophobic surfaces and a larger volume of accessible cellobiose units in cellulose III_I, as shown in Table 3. Cel7A seems to recognize the bulky, open structure of cellulose III_I, based on the

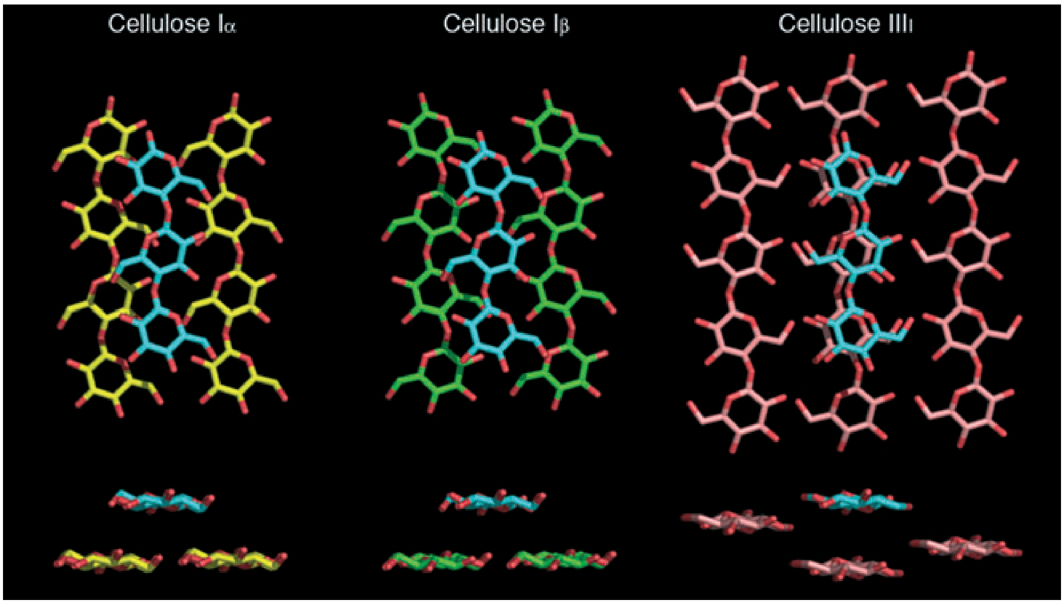


Fig. 6. Views from the hydrophobic surfaces (upper) and from the nonreducing end (lower) of cellulose I_α (left), cellulose I_β (middle), and cellulose III_I (right). The cellulose chains in the top layer are superimposed and colored cyan. The chains in the other layer are colored yellow (cellulose I_α), green (cellulose I_β), and magenta (cellulose III_I). The structures are based on the results reported by Nishiyama *et al.* [24,25] and Wada *et al.* [17].

Table 3. Distance between hydrophobic surfaces and volume occupied by a cellobiose unit in highly crystalline celluloses.

	Distance between hydrophobic surfaces (Å)	Volume of cellobiose unit (Å^3)
Cellulose I_α	3.91	333
Cellulose I_β	3.89	329
Cellulose III_I	4.27	347

order of hydrolysis (cellulose III_I > > cellulose I_α > cellulose I_β). In the previous study, we proposed that Cel7A distinguishes between the first and second layers of crystalline celluloses, as there is only small difference in packing density between cellulose I_α and I_β [14]. The enhanced hydrolysis of cellulose III_I observed in the present study indicates that the structural differences between cellulose I_α and I_β, i.e. differences of 0.02 Å in the distance of hydrophobic surfaces and 4 Å^3 in the volume of the cellobiose unit, are sufficient to explain the different hydrolytic characteristics. It is still uncertain how the crystalline polymorphic forms of cellulose affect the activity of cellobiohydrolase. However, we found that there was no significant difference in the hydrolytic rates of other fungal cellobiohydrolases when highly crystalline celluloses were used as substrates (data not shown). This result indicates that the rate-limiting step of hydrolysis is related to the crystalline form of cellulose, rather than the characteristics of the cellobiohydrolase. Generation of activated cellulose III_I, which is highly susceptible to cellobiohydrolase, seems to be the key to the effective hydrolysis of crystalline celluloses.

Experimental procedures

Preparations of crystalline celluloses

Cellulose I_α-rich and cellulose I_β (without ammonia treatment) samples were prepared from *Cladophora* sp. as described previously [14,26–28]. Cellulose III_I was prepared by supercritical ammonia treatment of *Cladophora* as described previously [29,30]. *Cladophora* samples treated with supercritical ammonia were further subjected to hydrothermal treatment in water at 160 °C for 30 min to generate cellulose $I_\beta{}'$ [27]. Scheme 1 shows an overview of the preparation of these samples.

Enzyme preparations and assays

Cel7A (formerly known as cellobiohydrolase I) from *Trichoderma viride* was purified from a commercial cellulase mixture, Meicelase (Meiji Seika Kaisha Co., Ltd, Tokyo, Japan) by three-step column chromatography as described previously [31,32]. The purity of the enzyme was confirmed by both electrophoresis and activity measurement. Crystalline cellulose samples (0.1% w/v) were incubated with various concentrations of enzyme (Abs_{280} = 0.04–1.6) in 1 mL 50 mM sodium acetate, pH 5.0, at 30 °C, and the reaction was terminated by centrifugation (15 000 ***g*** for 30 s). The absorbance at 280 nm of the supernatant was measured after the termination of the enzymatic reaction, and the concentration of free enzyme was determined using an absorption coefficient at 280 nm of 88 250 $M^{-1}\cdot cm^{-1}$ for *T. viride* Cel7A to estimate the amount of adsorbed Cel7A on crystalline celluloses [A; nmol·(mg cellulose)$^{-1}$] as described in the previous report [14]. To estimate cellobiose concentration in the supernatant, recombinant cellobiose dehydrogenase and cytochrome *c* were used as described previously [14,33].

Surface density analysis

The parameters required for surface density analysis, i.e. maximum high-affinity (A_1) and low-affinity (A_2) adsorptions [nmol·(mg cellulose)$^{-1}$], maximum adsorption ($A_{max} = A_1 + A_2$), constants for high-affinity (K_{ad1}) and low-affinity (K_{ad2}) adsorptions, surface density ($\rho = A/A_{max}$), rate of cellobiose production (v; μM·min^{-1}), and specific activity of adsorbed enzyme ($k = v/A$; min^{-1}), were calculated and estimated according to previous reports [14,34,35]. As a linear relationship was observed between ρ and ln k, the k value at $\rho \rightarrow 0$ (k_0) and the rate of reduction of k (B) were estimated using the following equation:

$$k = k_0 \exp(-B\rho) \quad (1)$$

It should be pointed out that we use Eqn (1) only for estimating k_0 and B for comparison of the hydrolytic rates for crystalline cellulose samples. We do not imply any physical interpretation of the equation or the constants, as they are empirical. The parameters were determined using DeltaGraph (version 5.5.1; SPSS Inc. and Red Rock Software, Inc.) and KaleidaGraph^{TM} (version 3.6.4 Synergy Software).

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